NOT MEASUREMENT SENSITIVE

MIL-PRF-32597
02 February 2018

PERFORMANCE SPECIFICATION

CLEANER, POLYMER AND POLYMER MATRIX COMPOSITE (PMC) MATERIAL, NON-AQUEOUS, HAP-FREE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a non-aqueous cleaner that is free of hazardous air pollutants (HAPs) and available in a low (no more than 25 grams per liter) volatile organic compound (VOC) and high VOC type for cleaning aircraft components (including structural metallic and fiber reinforced polymer matrix composite parts), ground support equipment, and other general cleaning applications (see 6.1).

1.2 Types. The cleaners are of the following types as specified (see 6.2).

Type I – Low-VOC (≤25 g/L)
Type II – High-VOC

1.2.2 Classes. The classes are as follows:

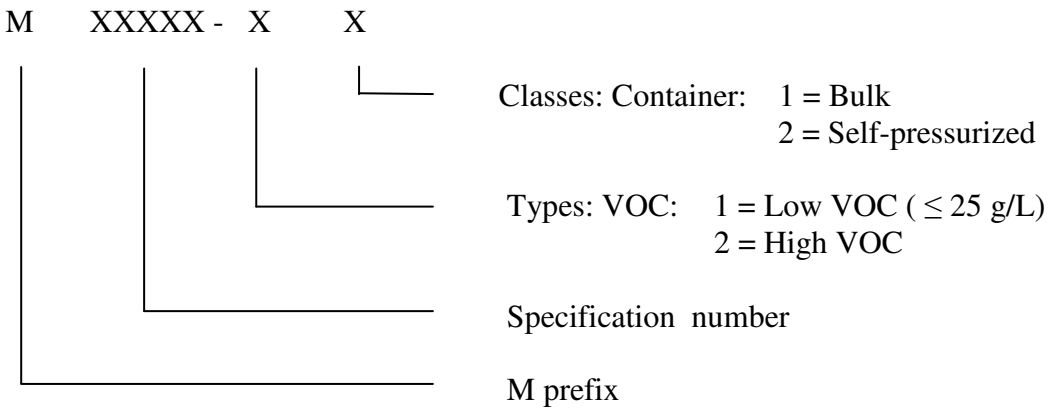
Class 1 - Bulk container
Class 2 - Self-pressurized container

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, (Code 4.1.2.2), Route 547, Mail Stop 120-3, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.dla.mil.

AMSC N/A FSC 6850

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1.3 Part or Identifying number (PIN). The PINs to be used for cleaners acquired to this specification are created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-10924	-	Grease, Automotive and Artillery
MIL-G-21164	-	Grease, Molybdenum Disulfide, for Low and High Temperatures, NATO Code Number G-353
MIL-DTL-81381/11	-	Wire, Electric, Fluorocarbon/polyimide Insulated, Medium Weight, Silver Coated Copper Conductor, 600 Volts, 200 Deg. C, Nominal 8.4 or 15.4 Mil Wall
MIL-PRF-81733	-	Sealing And Coating Compound, Corrosion Inhibitive
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil.)

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2.2.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 82	-	Protection of Stratospheric Ozone, Protection of Environment
40 CFR 401	-	General Provisions, Protection of Environment

(Copies of these documents are available from http://www.ecfr.gov.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Method 313	-	Determination of Volatile Organic Compounds (VOC) by Gas Chromatography/Mass Spectrometry (GC/MS)

(Copies of this document are available online at http://www.aqmd.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM C297/C297M-		Standard Test Method for Flatwise Tensile Strength of Sandwich Constructions
ASTM D56	-	Standard Test Method for Flash Point by Tag Closed Cup Tester (DoD Adopted)
ASTM D130	-	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test (DoD Adopted)
ASTM D891	-	Standard Test Methods for Specific Gravity, Apparent, of Liquid Industrial Chemicals (DoD Adopted)
ASTM D1002	-	Standard Test Method for Apparent Shear Strength of Single-Lap-Joint Adhesively Bonded Metal Specimens by Tension Loading (Metal-to-Metal)
ASTM D1296	-	Standard Test Method for Odor of Volatile Solvents and Diluents (DoD Adopted)
ASTM D1353	-	Standard Test Method for Nonvolatile Matter in Volatile Solvents for Use in Paint, Varnish, Lacquer, and Related Products (DoD Adopted)
ASTM D1613	-	Standard Test Method for Acidity in Volatile Solvents and Chemical Intermediates Used in Paint, Varnish, Lacquer, and Related Products (DoD Adopted)
ASTM D2240	-	Standard Test Method for Rubber Property-Durometer Hardness (DoD Adopted)

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ASTM D2344/ ASTM D2344M	-	Standard Test Method for Short-Beam Strength of Polymer Matrix Composite Materials and Their Laminates
ASTM D2879	-	Standard Test Method for Vapor Pressure-Temperature Relationship and Initial Decomposition Temperature of Liquids by Isoteniscope (DoD Adopted)
ASTM D3518/ ASTM D3518M	-	Standard Test Method for In-Plane Shear Response of Polymer Matrix Composite Materials by Tensile Test of a ±45° Laminate
ASTM D3755	-	Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electric Insulating Materials Under Direct-Voltage Stress
ASTM D5229/ ASTM D5229M	-	Standard Test Method for Moisture Absorption Properties and Equilibrium Conditioning of Polymer Matrix Composite Materials
ASTM D5868	-	Standard Test Method for Lap Shear Adhesion for Fiber Reinforced Plastics (FRP) Bonding
ASTM D6484/D6484M	-	Standard Test Method for Open-Hole Compressive Strength of Polymer Matrix Composite Laminates
ASTM D7264	-	Standard Test Method for Flexural Properties of Polymer Matrix Composite Materials
ASTM F483	-	Standard Test Method for Total Immersion Corrosion Test for Aircraft Maintenance Chemicals (DoD Adopted)
ASTM F484	-	Standard Test Method for Stress Crazing of Acrylic Plastics in Contact with Liquid or Semi-liquid Compounds (DoD Adopted)
ASTM F485	-	Standard Test Method for Effects of Cleaners on Unpainted Aircraft Surfaces
ASTM F502	-	Standard Test Method for Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces (DoD Adopted)
ASTM F519	-	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Plating/Coating Processes and Service Environments (DoD adopted)
ASTM F945	-	Standard Test Method for Stress-Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials (DoD Adopted)
ASTM F1110	-	Standard Test Method for Sandwich Corrosion Test (DoD Adopted)
ASTM F1111	-	Standard Test Method for Corrosion of Low-Embrittling Cadmium Plate by Aircraft Maintenance Chemicals (DoD Adopted)

(Copies of these documents are available from www.astm.org.)

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SAE INTERNATIONAL

SAE AMS 4377	Magnesium Alloy, Sheet and Plate, 3.0Al-1.0 Zn-0.20Mn (AZ31B-H24), Cold Rolled, Partially Annealed (DoD Adopted)
SAE AMS 4911	Titanium Alloy, Sheet, Strip, and Plate 6Al-4V Annealed (DoD Adopted)
SAE AMS 5040	Steel, Sheet and Strip 0.15 Carbon, Maximum Deep Forming Grade (DoD Adopted)
SAE AMS-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on (DoD Adopted)
SAE AMS-P-83310	Plastic Sheet, Polycarbonate, Transparent (DoD Adopted)
SAE AMS-QQ-A-250/4	Aluminum Alloy 2024, Plate And Sheet (DoD Adopted)
SAE AMS-QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet (DoD Adopted)
SAE AMS-S-8802	Sealing Compound, Fuel Resistant, Integral Fuel Tanks and Fuel Cell Cavities

(Copies of these documents are available from www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The cleaners furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material. The cleaners supplied under this specification shall be homogenous liquids that are clear (without turbidity), stable, and free of haze and suspended matter.

3.2.1 Composition. The cleaners shall be of the same composition and specific gravity as that of the qualification sample. If the cleaner contains any halogenated solvent, adequate acid acceptance stabilizers shall also be present. Any change in the formulation of a qualified product shall require requalification.

3.2.2 Prohibited materials. The cleaners shall contain no known or suspected human carcinogens (see 6.8.1), no hazardous air pollutants (see 6.8.2), no toxic pollutants (see 40 CFR), no ozone depleting substances (see 40 CFR 82), or any other hazardous substances.

3.3 Toxicity. The cleaners supplied under this specification shall have no adverse effect on the health of personnel when all precautions are followed for its usage in accordance with the applicable Safety Data Sheet (SDS) (see 4.5.16 and 6.7).

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3.4 Performance requirements. The cleaners shall meet the performance requirements listed in Table I.

TABLE I. Properties and test methods.

PROPERTY	REQUIREMENT		TEST METHOD
VOC content, grams/liter (maximum)	Type I ≤25	Type II No Limit	SCAQMD Method 313
Apparent specific gravity, 60/60 °F	No change from qualification sample		ASTM D891
Vapor pressure, mm Hg at 20 °C (maximum)	Type I 7	Type II 45	ASTM D2879
Flash point, °C (°F) (minimum)	60 (140)		ASTM D56
Nonvolatile residue, mg/100ml, (maximum)	5		ASTM D1353
Acidity (weight %)	0.02		ASTM D1613
Miscibility with water	Immiscible		4.5.1
Drying time, minutes (maximum)	50		4.5.2
Low temperature stability	No freezing and no separation		4.5.3
Sandwich corrosion (maximum)	Rating of 1		ASTM F1110
Immersion corrosion, mg/cm^2/day (maximum) Aluminum, Titanium, Steel Magnesium	 0.04 0.20		ASTM F483 and 4.5.4
Cadmium corrosion test, mg/cm^2/day (maximum)	0.20		ASTM F1111
Copper corrosion rating (maximum)	1b		ASTM D130 and 4.5.5
Effect on unpainted surfaces	No streaks or stains		ASTM F485
Hydrogen embrittlement	No failures in less than 150 hours when specimens are loaded to 45 percent of fracture strength and immersed in cleaner		ASTM F519 and 4.5.6
Titanium stress corrosion (examined with 500X magnification)	No cracking		ASTM F945 Method A
Effect on painted surfaces	No streaks, fading, blisters, or discoloration No softening >1 pencil hardness		ASTM F502

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TABLE I. Properties and test methods – Continued.

PROPERTY	REQUIREMENT		TEST METHOD
Effect on plastics Acrylic, type A & C Polycarbonate, SAE AMS-P-83310	 No crazing No crazing after 2 hrs at 2000 psi		ASTM F484
Effect on polyimide wire	No more insulation cracking than with distilled water and no subsequent dielectric breakdown or leakage		ASTM D3755 and 4.5.7
Effect on sealant	No change in Shore A hardness greater than ± 5 units		4.5.8
Effect on mechanical properties of fiber reinforced, polymer matrix composite	Data sets of baseline (unexposed) conditions and cleaner exposed conditions are not significantly different based on t-test or analysis of variance statistical methods		ASTM D2344/D2344M ASTM D7264 ASTM D3518/D3518M ASTM D6484/D6484
Effect on structural adhesive	Data sets of baseline (unexposed) conditions and cleaner exposed conditions are not significantly different based on t-test or analysis of variance statistical methods		ASTM D1002 ASTM D5868
Effect on structural honeycomb core	Data sets of baseline (unexposed) conditions and cleaner exposed conditions are not significantly different based on t-test or analysis of variance statistical methods		ASTM C297/C297M
Cleaning efficiency on MIL-PRF-83282 soil MIL-G-21164 soil MIL-PRF-10924 grease	Type I ≥ 95% ≥ 70% ≥ 85%	Type II ≥ 95% ≥ 70% ≥ 85%	4.5.12
Odor	Non-offensive, low intensity, non-residual		ASTM D1296 and 4.5.13

3.5 Storage stability. The cleaners, after storage in the original, unopened container for not less than one year at a temperature of 90 ±2 ºF (34 ±1 ºC), shall meet all the requirements of this specification. In addition, the containers shall exhibit no corrosion, distortion, leakage, or other evidence of incompatibility.

3.6 Propellant (Class 2). The propellant shall have zero Ozone Depleting Potential (ODP) and shall deliver a uniform spray of usable product. A propellant shall conform to Environmental Protection Agency (EPA) regulations for aircraft maintenance use of aerosol products.

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3.7 Aerosol dispenser (Class 2).

3.7.1 Composition. When visually examined in accordance with 4.5.15.1, the dispenser shall be of the aerosol type and shall have an on/off finger controlled pressure type of dispensing device to produce the spray. When not in use, the dispensing device shall be protected against activation by a snug-fitting protective cap. This cap shall provide a clearance of 0.125 inch (3.2 mm) over the dispensing device. The valve on the aerosol shall be of a type to allow for 360 degree actuation. In other words, the product can be dispensed in both an upright and upside down orientation. A 5 inch (127mm) hollow cylindrical plastic extension designed to fit into the dispensing device, for directing the spray into hard to reach places, shall be attached to the outside of the dispenser by tape, adhesive, or similar material.

3.7.2 Size. The dispenser shall be capable of containing 5 or 12 ounces (142 or 340 grams) of the cleaning compound (see 6.2), without exceeding 90 percent at 21 °C (70 °F) of the internal volume of the dispenser.

3.7. 3 Leakage. When tested as specified in 4.5.15.2, the pressurized dispenser shall not leak nor become distorted.

3.7. 4 Net contents. The average net contents per dispenser shall be not less than the amount specified for the size dispenser (see 6.2), when tested in accordance with 4.5.15.3.

3.7. 5 Precautionary markings. Each dispenser shall contain precautionary information, marked directly on the surface of the dispenser with marking material which is not affected by the cleaning compound. The warning shall be as follows:

CAUTION: Contents under pressure.
Do not puncture.
Use with adequate ventilation.
Keep away from direct sunlight, radiators, hot water, and other heat.
Avoid prolonged or repeated breathing of vapors or contact with skin.
Do not take internally.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection. The qualification inspection (see 6.3) shall consist of all tests and examinations of this specification. Failure of any examination or test shall be cause for rejection.

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4.3 Conformance inspection. Conformance inspection shall consist of the tests listed in Table II and the examinations listed in 4.3.2.

TABLE II. Conformance inspections.

TEST	REFERENCE
VOC	Table I
Apparent Specific Gravity	Table I
Vapor Pressure	Table I
Flash Point	Table I
Nonvolatile Residue	Table I
Acidity	Table I
Miscibility with Water	4.5.1
Drying Time	4.5.2
Low Temperature Stability	4.5.3
Sandwich Corrosion	Table I
Immersion Corrosion	4.5.4 & Table I
Cadmium Corrosion	Table I
Copper Corrosion	4.5.5 & Table I
Effect on Unpainted Surfaces	Table I
Hydrogen Embrittlement	4.5.6 & Table I
Titanium Stress Corrosion	Table I
Effect on Painted Surfaces	Table I
Effect on Plastics	Table I
Effect on Polyimide Wire	4.5.7 & Table I
Cleaning Efficiency	4.5.12
Odor	4.5.13 & Table I
Dispenser	4.5.15

4.3.1 Sampling for conformance tests. Samples for conformance tests shall be selected from each lot in accordance with MIL-STD-1916, verification level I. Samples for tests shall consist of unopened containers of cleaner selected at random from each lot. Containers shall only be opened when being tested. Any failure shall be cause for rejection of the lot.

4.3.2 Visual inspection of filled containers. Samples for visual inspection shall be selected at random from each lot in accordance with MIL-STD-1916, verification level I. The lot size for this inspection shall be the number of containers fully prepared for delivery. The selected samples shall be examined for container fill, completion of item identification, and chemical warnings. Any failure shall be cause for rejection of the lot.

4.4 Inspection conditions. Unless otherwise specified in the test method or paragraph, laboratory test conditions for all inspections and tests in this specification shall be 72 ±4 °F (22 ± 2 °C) and a relative humidity of 50 ±20 percent.

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4.5 Test methods. The tests of this specification shall be conducted in accordance with Table I and 4.5.1 through 4.5.16.

4.5.1 Miscibility with water. Twenty milliliters of the cleaner and 20 milliliters of distilled water shall be added to a 50 milliliter graduated cylinder and thoroughly shaken for 10 seconds. The mixture shall then be centrifuged (1000 rpm for 2 minutes) in a 50 milliliter graduated tube. The cleaner shall be considered immiscible if two clear phases separate cleanly with no volume change of either phase.

4.5.2 Drying time. To a pre-weighed aluminum weighing dish (all weights shall be to the nearest 0.0001 gram) 500 mg (+/- 10 mg) of cleaner shall be added. The weighing dish shall be 57 ±1 mm in diameter and no more than 1.5 grams in weight. The dish shall then be reweighed and immediately placed in a forced draft oven at 120 ±2 °F (49 ±1 °C). After 50 minutes, the dish shall be removed from the oven, cooled for 30 seconds by placing it on a clean metallic surface, and then immediately reweighed. If the cleaner remaining is no more than 0.1 percent of the original weight of cleaner used, the drying time shall be considered to be no more than 50 minutes.

4.5.3 Low temperature stability. Fifty milliliters of the cleaner shall be added to a test tube of suitable size and placed in a 0 ±2 °F (-18 ±1 °C) cold box for 24 hours. The tube shall be removed, immediately examined for any evidence of freezing, and allowed to return to room temperature, after which it shall be inverted five times and examined for separation.

4.5.4 Immersion corrosion. The immersion corrosion test shall be conducted in accordance with ASTM F483 (using the 7-day duration) on test panels constructed of the following materials:

a. Aluminum alloy 2024 (T3 temper), conforming to SAE AMS-QQ-A-250/4
b. Aluminum alloy 7075 (T6 temper), conforming to SAE AMS-QQ-A-250/12
c. Titanium alloy (6Al-4V), conforming to SAE AMS4911
d. Carbon steel (1020), conforming to SAE AMS5040
e. Magnesium alloy (AZ31B-H24), conforming to SAE AMS4377, chrome pickled to SAE AMS-M-3171, type VI

4.5.5 Copper corrosion. The copper corrosion test shall be conducted in accordance with ASTM D130 using a 24-hour soak in cleaner at 100 ±2 °F (38 ±1 °C).

4.5.6 Hydrogen embrittlement.

4.5.6.1 Preparation of test specimens. Hydrogen embrittlement characteristics shall be determined in accordance with ASTM F519, using cadmium-plated AISI4340, type 1a specimens. Four specimens shall be tested as follows. Plating shall be applied using treatment B and shall cover the notch and surfaces within 0.5 inch (13mm) of the notch; threaded surfaces shall not be plated. Plated specimens shall be baked in accordance with ASTM F519.

4.5.6.2 Test procedure. Each specimen shall be stressed by applying a load equivalent to 45 percent of notch fracture strength. The notch shall be immersed in the cleaner for the duration

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of the test. Discontinue the test if no fractures occur after 150 hours of immersion. If a failure occurs in one of four specimens tested, a second set of 4 specimens shall be evaluated. A minimum of 7 out of 8 specimens shall meet the 150 hour duration.

4.5.7 Effect on polyimide insulated wire. Approximately 24 inch (61cm) of wire conforming to MIL-DTL-81381/11 shall be formed into a coil. Place the coil in a 4 ounce (118ml) wide-mouth jar. Add to the jar sufficient cleaner to completely cover the coil of wire. Cap the jar and store it at room temperature for 14 days. Concurrently, perform this procedure using distilled water for comparison. At the end of the storage period, remove both coils and rinse thoroughly with tap water at room temperature. Suspend each coil and allow it to drain until completely dry. Each wire shall then be uncoiled and wrapped tightly on a 0.125 inch (3.2mm) mandrel noting the appearance of any cracks in the insulation. Each wire shall then be unwrapped and rewrapped in the opposite direction, again noting appearance of cracks. If the test wire shows no more cracks than the comparison wire, it shall be immersed in a 5 percent by weight sodium chloride solution and subjected to a one minute dielectric test of 2,500 volts (root mean square) without breakdown or leakage in accordance with ASTM D3755.

4.5.8 Effect on sealants. The effect of the cleaner on polysulfide and polythioether sealants shall be determined as specified in 4.5.8.1 and 4.5.8.2.

4.5.8.1 Preparation of test specimens. Sealants in accordance with AMS-S-8802, Type II, Class B and MIL-PRF-81733 Type II, Class 2, Grade B shall be mixed as specified by their respective manufacturers and processed to form sheets 0.125 inch (3.2mm) thick. Sealants shall be cured for 14 days at 77 ± 5°F (25 ± 3°C) at 50 percent humidity. These conditions shall be considered standard conditions. The sealing compound may be given an accelerated cure for 48 hours, minimum at standard conditions followed by 24 hours, minimum at 140 °F (60 °C). The cured sealants are the sheet stock from which the specimens for each sealant shall be cut.

4.5.8.2 Test procedure. The Shore A hardness of each sealant specimen shall be measured as specified in ASTM D2240 (stacking two specimens of the same sealant to make a 1/4 inch thickness) before exposing it to the cleaner. Two specimens of each sealant shall be immersed in cleaner at room temperature. After 30 minutes, specimens shall be removed, wiped dry and tested (as above) within 30 minutes for Shore A hardness.

4.5.9 Effect on mechanical properties of fiber reinforced polymer matrix composites. The effect of the cleaner on the mechanical properties of fiber reinforced polymer matrix composites shall be determined as specified in 4.5.9.1, 4.5.9.2, 4.5.9.3, 4.5.9.4. and 4.5.9.5.

4.5.9.1 Preparation of test specimens. Polymer matrix composite specimens for use in assessing the effects of the cleaner on mechanical properties shall be as described below:

a. For 4.5.9.2, 4.5.9.3, and 4.5.9.5, a balanced, quasi-isotropic laminate (minimum 16 plies total thickness) of unidirectional IM7/977-3 prepreg processed in an autoclave in accordance with material supplier recommendations.
b. For 4.5.9.4 a laminate of unidirectional IM7/977-3 prepreg stacked in accordance with ASTM D3518/D3518M Section 8.2.1 and processed in an autoclave in accordance with material supplier recommendations.

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4.5.9.2 Effect on short beam shear properties. A minimum of twenty Short Beam Shear test specimens shall be fabricated from the polymer matrix composite in accordance with ASTM D2344/D2344M. Test specimens shall be dried in accordance with ASTM D5229/D5229M, Procedure D. Ten dried specimens of the polymer matrix composite shall be immersed in the cleaner at room temperature for 14 days. After 14 days, specimens shall be removed from the cleaner, dried and tested in accordance with ASTM D2344/D2344M. Test results shall be compared to the short beam shear properties of the ten, baseline (unexposed) specimens using t-test or analysis of variance (ANOVA) statistical methods.

4.5.9.3 Effect on flexure properties. A minimum of twelve flexure test specimens shall be fabricated from the polymer matrix composite in accordance with ASTM D7264. Test specimens shall be dried in accordance with ASTM D5229/D5229M, Procedure D. Six dried specimens of the polymer matrix composite shall be immersed in the cleaner at room temperature for 14 days. After 14 days, specimens shall be removed from the cleaner, dried and tested in accordance with ASTM D7264. Test results shall be compared to the flexure properties of the six, baseline (unexposed) specimens using t-test or analysis of variance (ANOVA) statistical methods.

4.5.9.4 Effect on ± 45° tension properties. A minimum of twelve ±45° tension test specimens shall be fabricated from the polymer matrix composite in accordance with ASTM D3518/D3518M. Test specimens shall be dried in accordance with ASTM D5229/ASTM D5229M, Procedure D. Six dried specimens of the polymer matrix composite shall be immersed in the cleaner at room temperature for 14 days. After 14 days, specimens shall be removed from the cleaner, dried and tested in accordance with ASTM D3518/D3518M and ASTM C297/C297M. Test results shall be compared to the short beam shear properties of the six, baseline (unexposed) specimens using t-test or analysis of variance (ANOVA) statistical methods.

4.5.9.5 Effect on open hole compression properties. A minimum of twelve open hole compression test specimens shall be fabricated from the polymer matrix composite in accordance with ASTM D6484/D6484M. Test specimens shall be dried in accordance with ASTM D5229/ D5229M, Procedure D. Six dried specimens of the polymer matrix composite shall be immersed in the cleaner at room temperature for 14 days. After 14 days, specimens shall be removed from the cleaner, dried and tested in accordance with ASTM D6484/D6484M. Test results shall be compared to the ±45° tension properties of the six, baseline (unexposed) specimens using t-test or analysis of variance (ANOVA) statistical methods.

4.5.10 Effect on polymer based structural adhesives (paste and film). The effect of the cleaner on the mechanical properties of structural adhesives shall be determined as specified in 4.5.10.1. The test specimens shall be two balanced, quasi-isotropic laminates (minimum 16 plies total thickness) of unidirectional IM7/977-3 prepreg processed in an autoclave in accordance with material supplier recommendations bonded with 9394 structural adhesive, with the bonding accomplished in accordance with material supplier recommendations.

4.5.10.1 Effect on single lap shear properties. A minimum of twelve single lap shear test specimens shall be fabricated from the structural adhesive. The specimens shall be fabricated using materials described in 4.5.10.1 in accordance with ASTM D5868. Test specimens shall be

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dried in accordance with ASTM D5229/D5229M, Procedure D. Six dried specimens of the adhesive shall be immersed in the cleaner at room temperature for 14 days. After 14 days, specimens shall be removed from the cleaner, dried and tested in accordance with ASTM D5868. Test results shall be compared to the lap shear properties of the six, baseline (unexposed) specimens using t-test or analysis of variance (ANOVA) statistical methods.

4.5.11 Effect on polymer based structural honeycomb core. The effect of the cleaner on the mechanical properties of structural honeycomb and core shall be determined as specified in 4.5.11.1. The test specimens shall be a honeycomb sandwich comprising:

a. an aramid honeycomb core of 1 inch nominal thickness minimum, with minimum cell size of 0.125 inch nominal and minimum nominal density of 3.0 lb per cubic foot;
b. two facesheets of balanced, quasi-isotropic laminates (minimum 8 plies total thickness) of unidirectional IM7/977-3 prepreg processed in an autoclave in accordance with material supplier recommendations;
c. facesheets and core bonded with FM300 structural adhesive, with the bonding accomplished in accordance with material supplier recommendations.

4.5.11.1 Effect flatwise tensile properties of polymer based structural honeycomb core. A minimum of twelve flatwise tensile test specimens shall be fabricated from the structural honeycomb core described in 4.5.11 in accordance with ASTM C297/C297M. The test specimens shall be dried in accordance with ASTM D5229/D5229M, Procedure D. Six dried specimens of the core shall be immersed in the cleaner at room temperature for 14 days. After 14 days, specimens shall be removed from the cleaner, dried and tested in accordance with ASTM C297/C297M. Test results shall be compared to the flatwise tensile properties of the six, baseline (unexposed) specimens using t-test or analysis of variance (ANOVA) statistical methods.

4.5.12 Cleaning efficiency.

4.5.12.1 Preparation of test specimens. Stainless steel coupons 1 by 2 by 0.05 inch (25 by 50 by 1.3 mm) shall be polished with 240 grit aluminum oxide abrasive paper or cloth and solvent wiped with isopropyl alcohol. Coupons shall be weighed (weight = W_1), coated on one side with 20 - 25 mg of soil, then reweighed (weight = W_2). Soils to be tested shall be as follows:

a. MIL-G-21164 (with 10 percent by weight carbon black blended in using a grease working machine to achieve a smooth consistency)

b. MIL-PRF-83282 (with 10 percent by weight carbon black blended in using a spatula)

c. MIL-PRF-10924 grease without additions

4.5.12.2 Test procedure. Fresh solvent shall be used for each soil tested. Each test coupon shall be cyclically immersed and withdrawn from a 150ml beaker containing 100ml of the cleaner at a rate of 20 cycles per minute for 5 minutes. Each coupon shall then be dried for 10 minutes at 140 ±4 °F (60 ±2 °C), cooled to room temperature, and reweighed (weight = W_3).

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Cleaning efficiency for the cleaner shall be calculated as follows for each coupon:

$$\% \text{ Cleaning efficiency} = (W_2 - W_3) / (W_2 - W_1) \times 100$$

The test result for each soil shall be the average of three coupon cleaning efficiencies.

4.5.13 Odor evaluation. Odor shall be assessed by a three person panel in an odor-free area. No reference sample shall be used. Each assessor shall determine whether the odor is too offensive or intense to be used in the workplace and whether a residual odor exists after drying. The result shall be a consensus of the panel.

4.5.14 Storage stability. The cleaner, after storage in its original unopened container for not less than one year, shall be subjected to the tests necessary to meet the requirements specified in 3.5.

4.5.15 Aerosol dispenser.

4.5.15.1 Composition. The dispenser shall be visually examined to determine conformance to 3.7.1.

4.5.15.2 Leakage. Using a filled pressurized dispenser, the solvent shall be sprayed five times for three seconds with a two second pause between sprays. The dispenser shall then be submerged completely in water maintained at 54 ±1 °C (130 ±2 °F) for five minutes. The emission of bubbles indicates leakage. At the end of the five minutes, the dispenser shall be removed from the water and examined for conformance to 3.7.3.

4.5.15.3 Net contents. The net weight of the dispenser shall be determined by the difference between the gross weight and the tare weight. Results shall be in conformance with 3.7.4.

4.5.16 Health/hazard assessment (HHA). The cleaner shall be evaluated by the Navy and Marine Corps Public Health Center (NMCPHC), using the administrative HHA (see 6.7.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The materials covered by this specification are non-aqueous cleaners that are free of hazardous air pollutants and available in a low VOC (no more than 25 grams per liter) and high VOC types. The cleaners are intended for cleaning metallic, polymeric, and fiber reinforced polymer matrix composite materials and the adhesives and cores used in fabrication of advanced PMC-based aircraft structure. The cleaners are suitable for cleaning light soils, such as oils and hydraulic fluids, and heavy soils such as greases. Uses include cleaning of aircraft components and ground support equipment. The cleaner may be suitable for other cleaning applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Quantity.
d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-32597, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commander, Naval Air Warfare Center Aircraft Division, Code 4.3.4.2, Building 2188, Patuxent River, MD 20670-1908. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.
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6.3.1 Storage stability, inspection, and other information. In addition to the qualification test samples, the qualifying activity will request the manufacturer to submit to the qualification activity: (a) a certified test report showing that the material conforms to the requirements of this specification; (b) one copy of the SDS (see 6.7.1); (c) certification that the manufacturer's material meets the VOC requirement (see Table I).

6.3.1.1 Qualification inspection sample identification. Qualification inspection samples are to be forwarded to the laboratory designated in the letter of authorization from the qualifying activity (see 6.3) and identified as follows:

QUALIFICATION TEST SAMPLES
Specification MIL-PRF-32597, Type and Class (as applicable)
CLEANER, POLYMER AND POLYMER MATRIX COMPOSITE (PMC)

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MATERIAL, NON-AQUEOUS, HAP-FREE
Manufacturer's name and product number
Submitted by (name and date) for qualification testing in accordance with authorization (reference authorizing letter)

6.4 Retention of qualification. To retain qualification approval of products listed on the Qualified Products List (QPL), the manufacturer will be required to verify by certification to the qualifying activity that its product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification is in two-year intervals from the date of original qualification. The certification action is initiated by the qualifying activity.

6.5 Conformance rejection and retest. Failure in any conformance inspection will result in the rejection of the batch from which it was obtained and constitutes justification for removal from the qualified products list. Rejected material cannot be resubmitted for acceptance without written approval from the qualifying activity (see 6.3). The application for resubmission will contain all details concerning previous rejections and measures taken to correct these deficiencies.

6.6 Lot and batch formation. A lot consists of all of the cleaners manufactured at one time from one batch, forming part of one contract or order, and submitted for acceptance. A batch consists of all cleaners manufactured during one continuous operation.

6.7 Toxicity.

6.7.1 Safety Data Sheet (SDS). An SDS must be prepared and submitted in accordance with FED-STD-313 (obtain from https://assist.dla.mil). The SDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Questions pertinent to the effect(s) of these cleaners on the health of personnel using them should be referred by the procuring activity to the appropriate medical service, who will act as its adviser. Contracting officers will identify the activities requiring copies of the SDS.

6.7.2 Health/hazard assessment (HHA). The qualifying activity will request the Navy and Marine Corps Public Health Center (NMCPHC) to perform an HHA on the cleaner furnished. Prior to listing on the Qualified Products List, the results of the NMCPHC HHA must be deemed acceptable by the qualifying activity. The HHA is a review of the product based on information submitted by the manufacturer, to assess health hazards associated with the handling, application, and use of the product. Information on the HHA process can be found at http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx (BUMEDINST 6270.8A provides additional information/guidance pertaining to this process). Sufficient data to permit an HHA of the product is to be provided by the manufacturer to the Commanding Officer, Navy & Marine Corps Public Health Center, 620 John Paul Jones Circle, Suite 1100, Portsmouth, VA 20378-2103.

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6.8 Definitions.

6.8.1 Carcinogens. The Occupational Safety and Health Administration (OSHA) definition of carcinogens is those chemicals / processes appearing in lists 1, 2A, and 2B of the International Agency for Research on Cancer (IARC) 1/; substances known to be carcinogenic and occupational exposures associated with a technological process known to be carcinogenic by the National Toxicology Program (NTP) Report on Carcinogens (latest annual report) 2/; and OSHA regulated carcinogens.

1/ Copies of these volumes may be found in medical libraries or through the World Health Organization, 1211 Geneva 27, Switzerland.

2/ Copies may be obtained from the Public Health Service, National Toxicology Program, Public Information Office, P.O. Box 12233, MD B2-04, Research Triangle Park, NC 27709.

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6.8.2 Hazardous Air Pollutant (HAP). HAP is defined as any substance listed under Section 112 of the Clean Air Act or its modifications. The text of the Clean Air Act, listed pollutants and modifications are kept by the Environmental Protection Agency (EPA) and are accessible through the website: http://www.epa.gov.

6.8.3 Volatile organic compound (VOC). VOC is defined in 40 CFR Part 51.100.

6.9 Subject term (key word) listing.
Cleaning
Cleaning compound
Cleaning efficiency
Miscibility
Soil
Solvent

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 68
DLA – GS

Preparing activity:
Navy – AS
(Project 6850-2018-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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